Auditing Temporal-Context Leakage in Image-Based $PM_{2.5}$ Estimation: A Background-Conditioned Framework for Grouped Prediction

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Abstract

Random partitioning of overlapping image sequences can overstate generalization in environmental machine learning. We audit this failure mode and propose a background-conditioned framework combining an inference-time atmospheric or monitoring-network signal with learned visual and contextual refinement. On the mobile-roadside TRAQID dataset, a temporal–tabular model fell from R^2=0.950 under random windows, where 99.90% of test targets appeared in training context, to R^2=0.188 under zero-overlap complete-date evaluation. The framework was fitted independently to TRAQID and a six-site Taiwan network. TRAQID calibration-assisted prediction reduced future-period RMSE on 12 of 16 dates. On Taiwan, the complete model achieved R^2=0.706 across 54 later dates and R^2=0.702 under nested leave-one-site-out evaluation, with zero raw-frame overlap. These results connect

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frame-level leakage diagnosis to grouped, inference-realistic environmental prediction while making background-data requirements explicit.

Keywords: $PM_{2.5}$, temporal-context leakage, grouped validation, background conditioning, multimodal estimation

1. Introduction

Roadside $PM_{2.5}$ varies across regional pollution regimes, neighbourhood conditions and short-lived local sources. Fixed regulatory stations measure ambient concentration accurately but provide limited spatial coverage, while mobile sensing resolves route-level variation at the cost of substantial temporal dependence and changing scenes. Cameras potentially encode visibility, traffic mix and road state; atmospheric products encode broader aerosol and meteorological conditions. Learning from these sources is therefore attractive, but the evaluation unit must match the intended deployment question.

Temporal image studies often construct a sliding window around every target and then randomly split the resulting windows. Two adjacent seven-frame sequences share six frames. A model can consequently be tested on a target frame already seen as training context, even when target rows themselves are nominally disjoint. We use *temporal-context leakage* to denote train–test sharing of raw frames or overlapping sequence context; it is distinct from direct target leakage. Random cross-validation is known to underestimate error when dependence crosses partitions (Roberts et al., 2017), and leakage remains a major reproducibility issue in scientific machine learning (Kapoor and Narayanan, 2023).

This evaluation problem motivates the modelling reformulation in this paper. If a shuffled split allows a visual sequence model to reuse the regional pollution regime, route and adjacent frames, a high score does not show that the same feature system can identify local $PM_{2.5}$ conditions on a genuinely unseen date. We therefore separate the broad pollution state from the local departure and evaluate both only after complete dates or sites have been isolated. The paper is consequently both diagnostic and constructive: it first quantifies the contamination mechanism and then proposes a deployment-oriented multimodal framework for the harder grouped-prediction problem.

This paper uses two public image–pollution datasets with complementary group structure. TRAQID (Kathalkar et al., 2024) is a mobile roadside campaign spanning 20 dates and three seasons. The Taiwan subset released with the sequential-surveillance study of Wang et al. (Wang et al., 2026) contains six fixed-camera monitoring sites across 276 dates. Together they support four stages of investigation:

1. diagnose raw-frame contamination produced by randomly shuffled, overlapping driving sequences;
2. measure how the apparent result changes when complete collection dates are held out;
3. introduce background-conditioned multimodal local refinement as a response to the observed date shift; and
4. validate the framework under chronological future-period, leave-one-site-out and explicitly declared calibration-assisted tasks.

The model is formulated as

$$y_t = B_t + \Delta_t + \epsilon_t, \tag{1}$$

where y_t is roadside $PM_{2.5}$, B_t is an external atmospheric or local reference background, Δ_t is a learned local departure and ϵ_t contains measurement, alignment and unmodelled error. Traffic and road variables are predictors of conditions associated with Δ_t; they are not interpreted as chemically apportioned masses.

Equation 1 is a governing formulation rather than a claim that one fitted model transfers unchanged between datasets. TRAQID and Taiwan use separate estimators, feature sets and background sources because their sensing geometries and inference-time information differ. The shared question is whether an inference-time background, temporal imagery and structured context can be integrated into a useful complete system under grouped temporal or spatial evaluation.

The methodological contribution is a background-conditioned design principle, not a claim that one trained model transfers between campaigns. It couples an inference-time pollution background with learned local refinement and evaluates each dataset-specific instantiation under grouped shifts. The four contributions are:

1. a raw-frame audit of temporal-context leakage in random T=7 evaluation;
2. a background-conditioned formulation for local concentration refinement;
3. grouped temporal and spatial validation on two public datasets; and
4. robustness and calibration-assisted analyses that define the framework's operating limits.

The objective is a practical multimodal estimator rather than a camera-only monitor: broad pollution state is supplied by the atmospheric or monitoring background, while images and contextual variables encode local scene conditions. The same governing design is tested across mobile and fixed-camera settings, with zero-shot, network-assisted and calibration-assisted prediction reported separately.

2. Related Work

Image-based air-quality estimation uses atmospheric attenuation, colour and scene context as indirect aerosol cues. Luo et al. combined convolutional image features with gradient boosting and weather/time variables (Luo et al., 2020). Wang et al. used surveillance imagery with a CNN–LSTM and reported R^2=0.94 for $PM_{2.5}$ under random two-fold sequence cross-validation (Wang et al., 2024). Liu et al. reported strong performance from a repeated fixed traffic-camera scene paired with a reference monitor (Liu et al., 2024). PM25Vision broadened visual benchmarking to geographically distributed street imagery, but uses date-level AQI labels rather than instantaneous roadside concentration (Han, 2026).

These results are not directly rankable: scene stability, label construction, concentration range, aggregation interval and split unit differ. The present study therefore compares protocols within TRAQID rather than treating published R^2 values as a leaderboard. The central distinction is whether evaluation measures interpolation among related sequences or prediction on a complete unseen date.

Regional products provide a second source of information. MERRA-2 assimilates observations into a global meteorological and aerosol reanalysis (Gelaro et al., 2017); CAMS provides global atmospheric-composition analyses and forecasts (Inness et al., 2019). Their pixels are too coarse to replace a roadside measurement directly, but they can anchor the broad background in Eq. 1. A nearby station can provide a more local anchor, at the cost of changing the task to reference-assisted estimation.

This formulation sits within a longer exposure-modelling literature rather than replacing it. Land-use regression uses traffic, land use and geographic covariates to resolve intra-urban contrasts (Hoek et al., 2008); statistical downscalers calibrate coarse numerical-model output against observations (Berrocal et al., 2010); and modern exposure models combine monitors, satellite retrievals, chemical-transport output, meteorology and land use (Di et al., 2019; Wei et al., 2020). Those studies also show why random sample-level validation is not equivalent to spatial or temporal transfer. Here, imagery is tested as an incremental local predictor against explicit atmospheric and monitoring-network baselines, not as a substitute for those established sources.

3. Data

3.1. TRAQID campaign

TRAQID is a public mobile traffic-related air-quality image dataset collected across Hyderabad and Secunderabad, India, from October 2022 to July 2024 (Kathalkar et al., 2024). The platform recorded paired front and rear images and a custom AQ Node approximately every five seconds during more than 70 hours of driving. A Nova SDS011 measured $PM_{2.5}$ and PM_{10}; a BME280 measured temperature and relative humidity. The low-cost particle measurements were calibrated against an Aeroqual S500 reference instrument in the source study.

After alignment and T=7 construction, 26,558 sequences remained across 20 dates: six monsoon dates, ten winter dates and four summer dates. Roadside $PM_{2.5}$ ranged from 12.92 to 400.86 $\mu g\, m^{-3}$, with mean 66.58 and median 52.43 $\mu g\, m^{-3}$. Figure 1 shows that both sequence count and target distribution changed markedly by date. Rows are therefore numerous but are not independent temporal replicates; the effective diversity is closer to 20 date clusters than 26,558 independent samples.

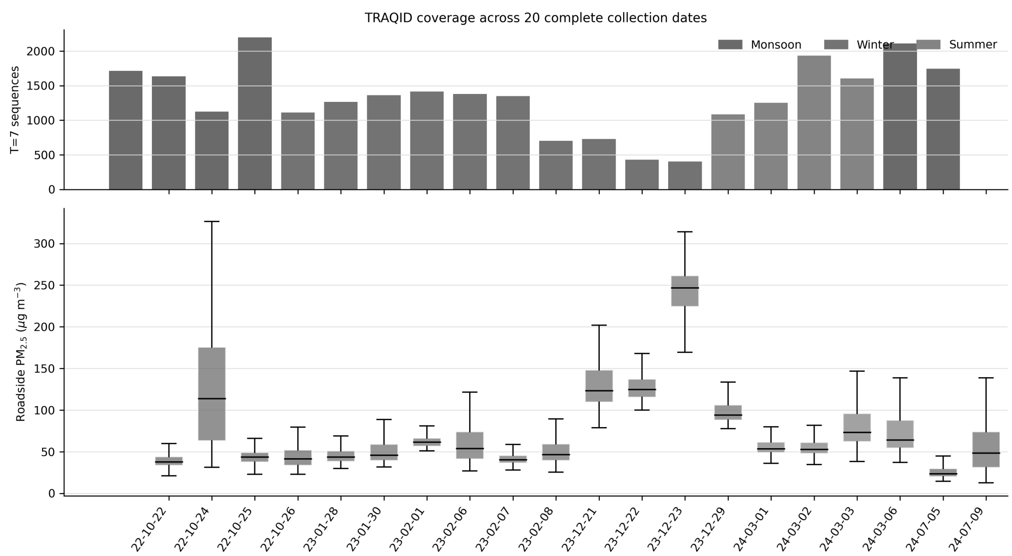


Figure 1: TRAQID date coverage and roadside $PM_{2.5}$ distributions. Shading denote the season labels used in the campaign-level analysis. Box plots omit fliers for legibility but all values are retained in modelling.

3.2. Image, traffic and road variables

Front and rear frames were represented with ImageNet ResNet50 global-average pooling features (He et al., 2016). Seven-frame sequences were aggregated or encoded temporally. Structured visual features covered traffic counts, class mix, confidence, bounding-box occupancy, near-field traffic and spatial subregions for cars, motorcycles, auto-rickshaws, buses, trucks, bicycles and people. Road-region variables described area, brightness, saturation, contrast, shadow, glare, dry/brown pixels, edges, texture and a haze proxy. The engineered T=7 table contained 102 source feature families summarized by latest value, mean, standard deviation, minimum, maximum and change.

3.3. Atmospheric and reference context

Twenty-two atmospheric inputs were aligned by timestamp: MERRA-2 aerosol components and derived $PM_{2.5}$ (Gelaro et al., 2017; Google Earth Engine Data Catalog, 2026c); CAMS PM_1, $PM_{2.5}$, PM_{10} and aerosol optical-depth fields (Inness et al., 2019; Google Earth Engine Data Catalog, 2026a); and ERA5-Land temperature, humidity, wind, pressure, precipitation and solar radiation (Google Earth Engine Data Catalog, 2026b). Atmospheric values were not fitted to the TRAQID target.

Historical OpenAQ observations were searched within 50 km of the campaign centre (OpenAQ, 2026). The retrieval artifact records 23 $PM_{2.5}$ sensors at 13 locations. After temporal and quality filtering, usable contemporaneous history came from OpenAQ location 8557 ("Hyderabad"), sensor 24857, supplied by AirNow at 17.38405°N, 78.45636°E. OpenAQ did not expose an instrument model for this record. The site was approximately 4.97 km from the campaign centre. Each TRAQID timestamp was paired to the nearest hourly value only when the absolute gap was at most 90 minutes; longer gaps were left missing, not interpolated. Complete coverage was obtained for 16 of 20 dates and 20,875 of 26,678 raw rows. The four uncovered dates (24 October 2022 and 6–8 February 2023) were excluded from reference-assisted comparisons. The corresponding T=7 table contained 20,779 sequences. These results are reference-assisted, not camera-only predictions.

3.4. Taiwan fixed-camera monitoring network

The second evaluation uses the Taiwan portion of the public sequential surveillance-image air-quality dataset released by Wang et al. (Wang et al., 2026). It contains hourly fixed-camera images and co-timed $PM_{2.5}$ observations at six government monitoring locations in southern Kaohsiung. The source publication identifies the locations as Qiaotou, Renwu, Fengshan, Daliao, Linyuan and Xiaogang, spanning inland urban/suburban settings and the coastal industrial belt. The redistributed folders use codes 48, 49, 50, 51, 52 and 58 in that order; these are treated as dataset codes, not assumed to be current Ministry of Environment station identifiers. After continuity-constrained T=7 construction, 27,084 sequences remained across 276 dates. Measured $PM_{2.5}$ ranged from 2 to 106 $\mu g\,m^{-3}$ (mean 26.28; median 26.00).

The source data cover July 2019 through April 2020 and contain 1,280×720 surveillance images paired with hourly government-station pollution records. The processed manifest retained 28,611 rows with no missing $PM_{2.5}$ labels; availability varied from 3,743 to 5,952 rows and 169 to 273 dates per site. Table A.10 reports the site names and retained coverage. The redistributed files do not embed latitude/longitude, pairwise distance, station address or instrument-model fields. We therefore report the verified district-level locations from the source publication without reverse-engineering coordinates or monitor hardware from the numeric folder codes.

The 276 dates were ordered chronologically before sequence construction. The earliest 162 dates formed training, the next 52 validation and the final 54 test, producing 13,154, 6,750 and 7,180 sequences respectively. No raw image frame occurred in more than one partition. At each target timestamp, the monitoring-network background was the median exactly contemporaneous (same clock-hour; zero synchronization tolerance) $PM_{2.5}$ across the other available sites. The target site's own measurement was excluded by construction; consequently, the background is a network-assisted input rather than target leakage. This task estimates one site's concentration given images at that site and measurements elsewhere in the network. Between one and five supporting sites were available per retained sequence (mean 4.04, median five); on the 7,180-sequence test partition, 5,892 sequences used all five support sites and none used fewer than two.

For a complementary spatial test, six outer folds held out one complete site at a time. Within every outer fold, the image-correction weight was selected by cross-validation over training sites only. All six sites were therefore scored once as held-out spatial groups. This is network-assisted spatial transfer within southern Kaohsiung, not sensor-free transfer to a new city.

4. Methods

4.1. Unified background-conditioned formulation

Figure 2 separates regional or network-scale background from local estimation. A timestamp and location select an inference-time background $B(t, s)$. A T=7 visual encoder estimates a local visual departure, while a structured branch estimates departure from atmospheric, temporal, traffic, road or site context available in that dataset. Validation data select the admissible fusion; the background is added once to reconstruct the final concentration. Predictors are covariates of one concentration and are not interpreted as separately additive source masses.

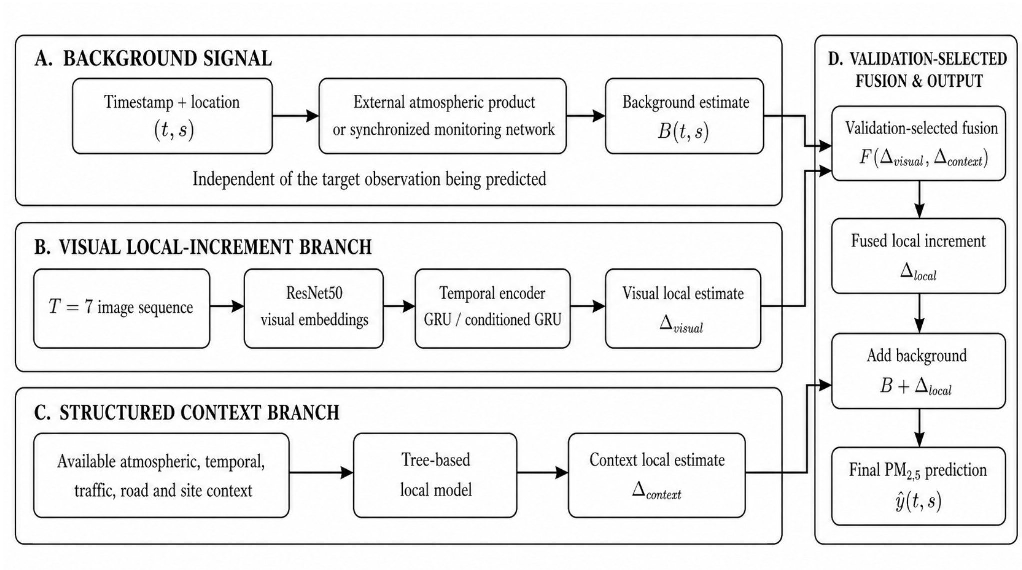


Figure 2: Shared background-conditioned architecture. TRAQID instantiates B with MERRA-2, CAMS or a nearby monitoring station and combines temporal and tabular local estimates. Taiwan instantiates B with the synchronized median of other sites, uses a cross-fitted context model and adds a bounded, validation-shrunk conditioned-image correction. The governing decomposition is shared; feature availability and the fitted fusion are dataset-specific.

The inference assumptions are deliberately explicit. In TRAQID, MERRA-2 and CAMS are coarse atmospheric products; the OpenAQ variant additionally requires a contemporaneous nearby monitor and is therefore reference-assisted. In Taiwan, prediction at site s requires simultaneous observations from the other network sites, while site s is excluded.

These inputs are legitimate covariates only when they would be available in the intended deployment. They must not be compared as though they represented the same sensing burden.

Table 1: Dataset-specific instantiations of the common decomposition. No fitted weights are shared across datasets.

Dataset	Background B	Learned local component	Required at inference	Excluded information
TRAQID atmospheric	MERRA-2 or CAMS	Mobile image sequence + structured context	Timestamp, atmospheric products, scene covariates	Held-out-date targets
TRAQID assisted	Nearby OpenAQ monitor	Reference context + local covariates	Same monitor throughout the drive	Future 90% targets
Taiwan network	Median of other sites	Context model + conditioned image residual	Other sites at target hour + target-site image	Target site's own observation

4.2. Validation protocols

Table 2 separates the evaluation questions. Random T=7 windows retain the published-style diagnostic but do not measure deployment. LODO holds one complete date out, fits preprocessing and estimators on the other dates, and concatenates each untouched outer-test prediction for pooled metrics. The calibration protocol begins from these frozen predictions, uses the earliest 10% of the held-out date as adaptation data, removes every future sequence sharing any raw row with adaptation, and evaluates the remainder.

Table 2: Evaluation protocols and their permitted interpretation.

Protocol	Split unit	Direct frame overlap	Interpretation
Random overlapping T=7	Sequence after window construction	Severe	Same-distribution diagnostic only
20-date LODO	Complete collection date	None	Zero-shot cross-date evaluation
16-date reference LODO	Complete reference-covered date	None	Zero-shot, reference-assisted evaluation
10% adaptation / future 90%	Chronological within held-out date, exact raw-row purge	None	Calibration-assisted future-period prediction
Taiwan chronological 60/20/20	Complete dates assigned before T=7 construction	None	Retrospective future-period evaluation
Taiwan nested LOSO	Complete monitoring site; inner training-site CV	None	Network-assisted spatial transfer
Oracle corrections	Entire evaluated date	Uses all test targets	Diagnostic ceiling only

4.3. Random-window architecture and overlap audit

The random benchmark used an external MERRA-2 background and two local branches. A ResNet50–GRU T=7 temporal branch predicted the local increment and received cross-fitted Random Forest residual correction (He et al., 2016; Cho et al., 2014; Breiman, 2001). An ExtraTrees branch used 631 multiscale atmospheric, weather, traffic and road predictors (Geurts et al., 2006). A validation-trained conditional gate combined the branch predictions. Train, validation and test targets were disjoint, but the raw frame identifiers composing every sequence were audited across partitions.

4.4. Complete-date background-decomposition models

For full 20-date LODO, MERRA-2 supplied B_t. Local ExtraTrees models compared nonvisual context, nonvisual plus images, nonvisual plus structured visual and all modalities. Image features were reduced by train-only randomized PCA. All imputation, scaling, PCA and model fitting were restricted to outer-training dates.

For the 16 reference-covered dates, the OpenAQ measurement was forced as B_t. Reference-context variables included the current level, T=7 mean, standard deviation, change and slope, plus contrasts against CAMS and MERRA-2. The strongest zero-shot branch used reference context, meteorology, time and atmospheric variables. Images and structured visual variables were retained as matched ablations rather than assumed to be beneficial.

Both TRAQID grouped runners used date-balanced sample weights and median imputation with missingness indicators. Local ExtraTrees models used 300 trees, minimum leaf size 8 and `max_features`=0.7. ImageNet ResNet50 frame embeddings were reduced by train-only randomized PCA to at most 48 components; latest, mean, standard deviation and first-to-last change over T=7 were then concatenated. The reference-assisted fusion used four inner date-grouped folds and a simplex step of 0.05. The chronological meta-calibrator standardized its single early-residual feature and selected ridge $\alpha \in \{0.01, 0.1, 1, 10, 100\}$ by leave-one-date-out cross-validation among the other 15 dates. All random seeds were 42 unless a deterministic fold offset was recorded by the runner.

4.5. Taiwan monitoring-network and conditioned-image model

For target site s, the background $B(t, s)$ was the contemporaneous median of the other available sites. A six-variable context vector contained $B(t, s)$, the contributing-site count and cyclic hour and day-of-year terms. ExtraTrees predicted the local increment $y - B$. Training predictions were cross-fitted by site before constructing the neural correction target, preventing the image branch from learning an in-sample residual of the context model.

Each T=7 image sequence was represented by ResNet50 embeddings. A context encoder generated feature-wise linear modulation parameters for a one-layer GRU with hidden width 128. The network predicted the remaining correction after the context branch; a hyperbolic tangent bounded its magnitude at 15 $\mu\mathrm{g\,m}^{-3}$. The final estimator was

$$\widehat{y}(t, s) = B(t, s) + \widehat{\Delta}_{\mathrm{ctx}}(t, s) + \alpha\, \delta_{\mathrm{img}}(t, s). \quad (2)$$

For the chronological experiment, checkpoint and $\alpha = 0.30$ were selected on the 52-date validation partition and frozen before the 54-date test was scored. For outer leave-one-site-out evaluation, $\alpha \in [0, 0.30]$ in steps of 0.02

was selected separately using cross-validation among outer-training sites; the held-out site's targets were never used for selection.

The context ExtraTrees model used 400 trees, minimum leaf size 10 and `max_features`=0.8. The conditioned image network used a 256-dimensional projection of frozen ResNet50 embeddings, a one-layer GRU with hidden width 128, dropout 0.3 and a correction bound of 15 μg m^{-3}. It was optimized with AdamW (learning rate 10^{-4}, weight decay 10^{-4}), Huber loss with $\delta = 3$, correction penalty 0.005, batch size 64, at most 80 epochs and patience 12. Checkpoint and correction weight were selected exclusively on the declared validation groups. Appendix A records exact artifacts and selection status.

After the frozen chronological test had been inspected, an exploratory network ablation compared learned spatial and lagged spatiotemporal backgrounds and a validation-selected nonnegative ensemble. Its split controls remain valid, but its test values are labelled exploratory and are not used as confirmatory headline results.

4.6. Chronological calibration-assisted model

For the TRAQID calibration-assisted experiment, the earliest 10% of each held-out date was assigned to adaptation; ties at the boundary timestamp remained together. Every later sequence whose seven raw rows intersected adaptation was purged. This removed six boundary sequences per date (96 total), leaving 18,554 future sequences after 2,129 adaptation sequences.

Let e_d^{early} be the median residual $y - \hat{y}$ in the adaptation segment and e_d^{future} the mean residual in the later segment. For target date d, a ridge mapping

$$\hat{e}_d^{\text{future}} = f_{-d}\left(e_d^{\text{early}}\right) \tag{3}$$

was fitted using the other 15 dates only. Ridge strength was selected by leave-one-date-out cross-validation inside those 15 dates. The calibrated prediction was

$$\hat{y}^{\text{cal}}_{d,t} = \hat{y}_{d,t} + \hat{e}^{\text{future}}_{d}. \tag{4}$$

No target from the future 90% of date d entered f_{-d}, its hyperparameter selection or the frozen base model. Direct mean, median and affine corrections fitted only on the first 10% were included as controls.

4.7. Metrics and uncertainty

We report MAE, RMSE, bias and pooled R^2 after concatenating outer-test predictions. Because within-date variance differs markedly, per-date R^2 can be unstable; rather than omit it, we report its mean, median and number of positive dates alongside macro-averaged date MAE/RMSE. Date-centred R^2 subtracts the observed and predicted mean within each date before pooling and therefore measures within-date shape independently of between-date offsets. A deterministic non-overlapping sensitivity retains every seventh target in time order within each date, so retained T=7 windows do not share raw frames. TRAQID date-cluster uncertainty resampled the 16 complete dates with replacement 5,000 times, retaining every row within a sampled date. Taiwan uncertainty used 20,000 paired resamples of the 54 chronological test dates and, separately, 20,000 paired resamples of the six outer test sites. Percentile intervals therefore measure sensitivity to the relevant grouped units rather than treating dense sequences as independent samples. All analyses used fixed seed 42 and scikit-learn (Pedregosa et al., 2011).

The primary endpoint for TRAQID was matched future-90% RMSE after calibration, compared with the uncalibrated prediction on exactly the same sequences. The primary Taiwan endpoint was chronological-test RMSE relative to the prespecified contemporaneous other-site median. Pooled R^2, grouped R^2 and correlation are secondary descriptors. This ordering was fixed for the manuscript revision to avoid selecting the most favourable metric after inspection.

The study was developed iteratively. The random-window reproduction, backbone comparisons and several background variants were exploratory. The grouped protocols, exclusion of target-site measurements, chronological partitions and primary matched comparisons were then evaluated from preserved split manifests and prediction files. Appendix A distinguishes diagnostic, grouped, assisted and post-hoc results. A future prospective study should freeze the full analysis plan before acquiring target labels.

5. Results

The results follow the study logic. Section 5.1 diagnoses the shuffled-window failure mode. Sections 5.2–5.4 then evaluate the proposed background-conditioned framework under chronological, held-out-site and complete-date shifts, and Section 5.5 tests the explicitly assisted prediction setting. Component-level comparisons are retained in the appendix so that the main sequence remains diagnosis, reformulation and grouped validation.

5.1. Protocol audit separates interpolation from grouped prediction

The exact random-window conditional architecture achieved MAE 5.36 $\mu\mathrm{g\,m^{-3}}$, RMSE 10.29 $\mu\mathrm{g\,m^{-3}}$ and pooled R^2=0.950 on 3,983 test sequences. However, 24,546 of 26,678 raw rows occurred in more than one partition, 18,159 raw rows overlapped train and test, and 3,979 of 3,983 test targets

(99.90%) had already appeared as training context. The result demonstrates reproducible interpolation under the shuffled-window protocol, not future-date prediction.

The same audit applied to random two-fold sequence cross-validation—the evaluation pattern used by representative surveillance-image studies—found that 74.81% of test sequences shared at least six of seven raw frames with a training sequence and only 0.01% had zero overlap. This does not establish that every published image-based result is contaminated; it establishes that overlapping-window protocols are structurally vulnerable unless raw-frame separation is verified explicitly.

Table 3: Frame-overlap audit for representative TRAQID protocols.

Protocol	Eval n	Mean max overlap	Share $\geq 6/7$	Share zero
Random 70/15/15	3,983	5.899	90.91%	0.00%
Random two-fold	13,279	5.667	74.81%	0.01%
Time-balanced purged	6,691	0.000	0.00%	100.00%
Complete-date	5,462	0.000	0.00%	100.00%
Chronological 10% adaptation	18,554	0.000	0.00%	100.00%

5.2. Grouped multimodal estimation across dates and sites

The Taiwan experiment supplied a stronger fair test because it contained 276 dates and six synchronized sites. Figure 3 documents the partition audit. Dates were assigned before windows were constructed, raw-frame overlap across train, validation and test was zero, and a target site's reading never contributed to its own background. The test partition was used once for scoring after the checkpoint and image weight had been fixed on validation.

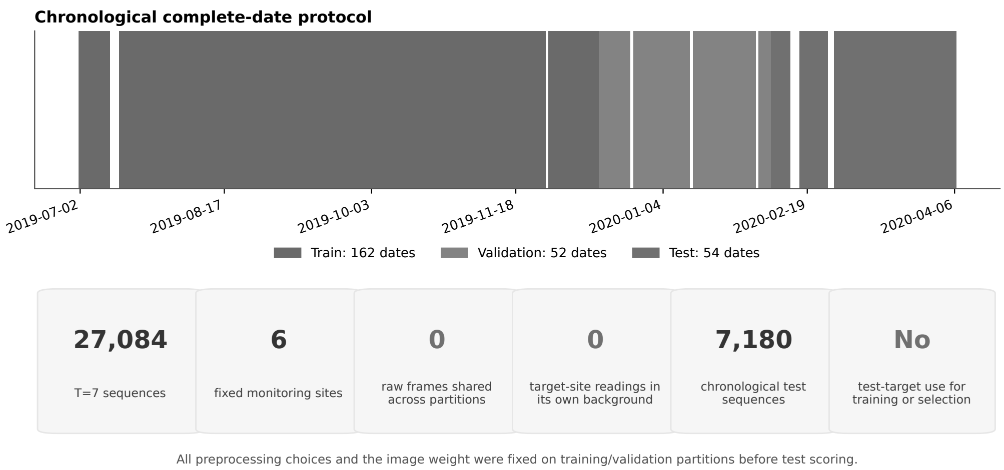


Figure 3: Taiwan chronological protocol and leakage audit. The earliest 162 dates train the model, 52 subsequent dates select it and the final 54 dates provide 7,180 test sequences. All reported counts are generated from the frozen sequence manifest and split audit.

The primary Taiwan endpoint was chronological-test RMSE relative to the prespecified other-site median. On the future period, the synchronized network background alone obtained MAE 4.70 $\mu g\, m^{-3}$, RMSE 6.61 $\mu g\, m^{-3}$ and R^2=0.690. The background-plus-context-plus-conditioned-image model improved these to 4.58, 6.44 and 0.706, respectively (Table 5). Its Pearson and Spearman correlations were 0.846 and 0.868. Date-block resampling estimated a median RMSE reduction of 0.172 $\mu g\, m^{-3}$ (95% interval 0.082–0.254) and an R^2 gain of 0.016 (0.007–0.026); 49 of 54 dates had positive within-date R^2 and RMSE decreased on 41 dates.

Transparent network baselines clarify where this skill originates (Table 4). A training-only target-site/hour climatology failed on the future period (R^2=-0.116), and delaying the other-site median by one hour reduced R^2 to 0.647. Contemporaneous other-site mean and median baselines reached 0.696 and

0.690. Relative to the median, the conditioned-image estimator improved MAE by 0.124 $\mu\mathrm{g\,m^{-3}}$, RMSE by 0.172 $\mu\mathrm{g\,m^{-3}}$ and R^2 by 0.016 in 5,000 paired complete-date resamples; all three 95% intervals excluded zero. Removing any one support station changed median-background RMSE from 6.61 to 6.68–6.81 $\mu\mathrm{g\,m^{-3}}$, showing moderate rather than catastrophic single-station sensitivity.

Table 4: Transparent baselines on the identical frozen 54-date Taiwan test partition. The target site's contemporaneous observation is absent from every network predictor.

Model	MAE	RMSE	Pooled R^2	Coverage
Training mean	10.035	12.717	-0.146	100.0%
Target-site/hour climatology	9.923	12.550	-0.116	100.0%
Other-site median, one-hour lag	5.050	7.061	0.647	99.85%
Other-site contemporaneous median	4.704	6.613	0.690	100.0%
Other-site contemporaneous mean	4.718	6.553	0.696	100.0%
Frozen conditioned-image estimator	**4.580**	**6.442**	**0.706**	100.0%

The network-assisted spatial result was consistent. Nested leave-one-site-out evaluation scored 27,019 sequences and obtained pooled MAE 4.82 $\mu\mathrm{g\,m^{-3}}$, RMSE 6.46 $\mu\mathrm{g\,m^{-3}}$ and R^2=0.702. Every held-out site had positive R^2 (0.534–0.778), with mean and median site-wise R^2 of 0.692 and 0.709. Fusion weights were selected only within the training sites, so the complete framework was tested without using the outer site's labels. Detailed component and weight sensitivity is reported in Appendix F; the principal result is that the combined estimator retained positive skill under both chronological and held-out-site evaluation. The other-site network already carried most of the predictive signal; conditioned imagery provided a small, statistically measurable refinement rather than replacing that network.

Table 5: Taiwan grouped framework evaluation. Chronological rows share one frozen future-period test set. LOSO rows concatenate six outer test sites; fusion is selected only by inner training-site cross-validation. Detailed component rows are reported in Appendix F.

Protocol	Model	n	MAE	RMSE	Pooled R^2	Status
Chronological	Other-site network background	7,180	4.704	6.613	0.690	Frozen baseline
Chronological	Background + context + conditioned image	7,180	**4.580**	**6.442**	**0.706**	Frozen test
Nested LOSO	Other-site network background	27,019	4.868	6.604	0.689	Outer test
Nested LOSO	Conservative conditioned image	27,019	**4.818**	**6.460**	**0.702**	Inner-CV selected

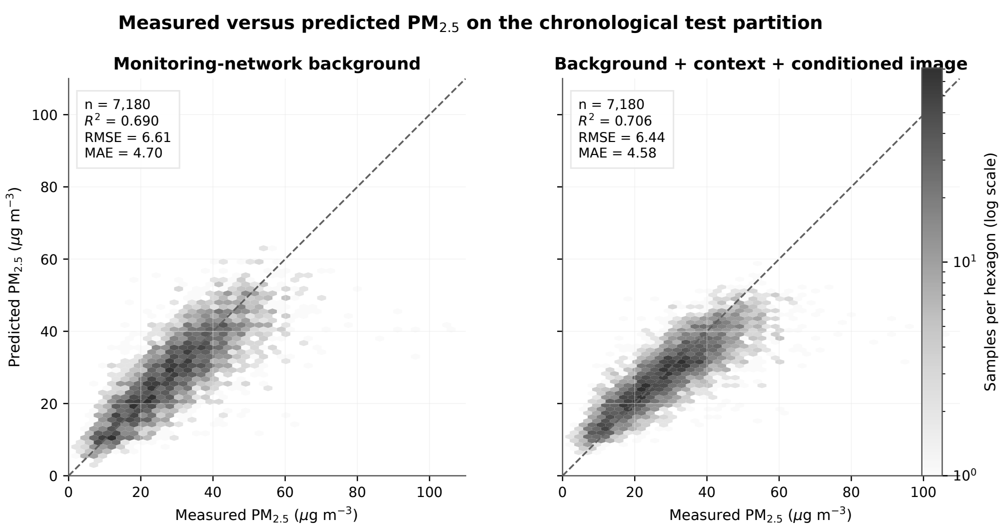


Figure 4: Measured versus predicted Taiwan $PM_{2.5}$ on the chronological 54-date test period. Both panels use identical 7,180 sequences and axes. The conditioned model reduces aggregate error but still underestimates sparse high-concentration observations.

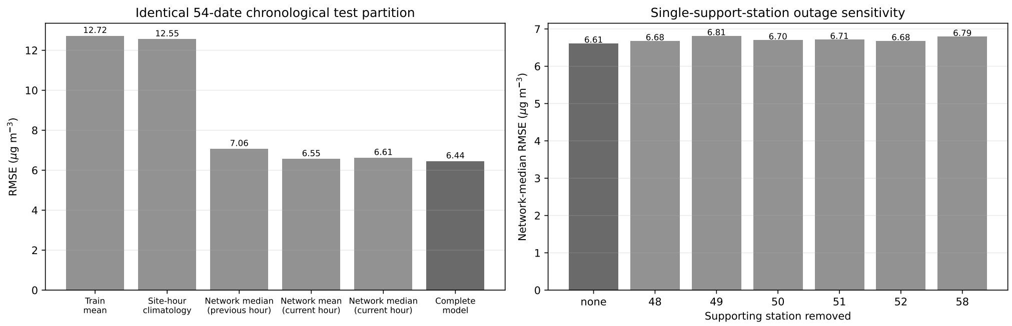


Figure 5: Monitoring-network baseline and reliability audit. Left: RMSE on the identical 54-date chronological test partition. Right: network-median RMSE after removing one support site at a time.

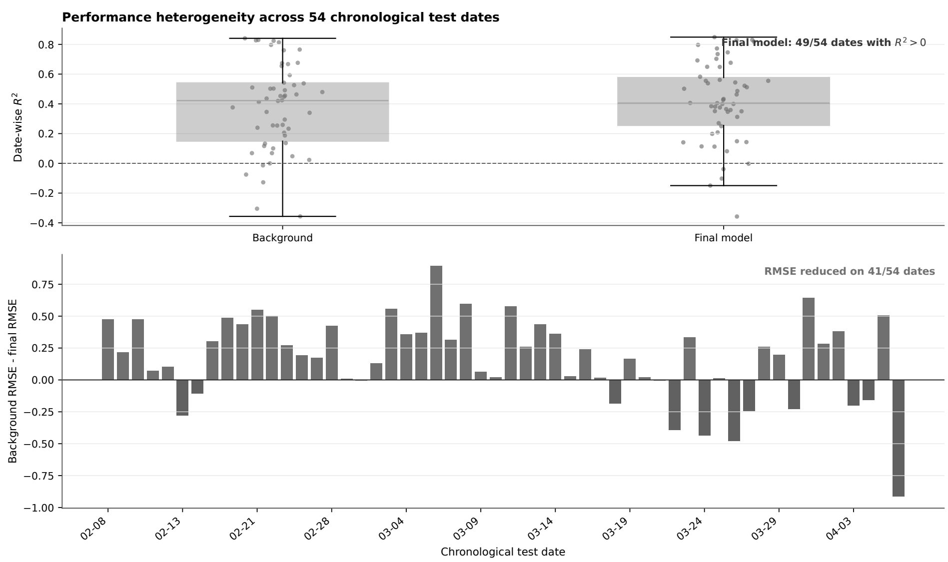


Figure 6: Taiwan complete-date robustness on the chronological test. The visual correction does not improve every date.

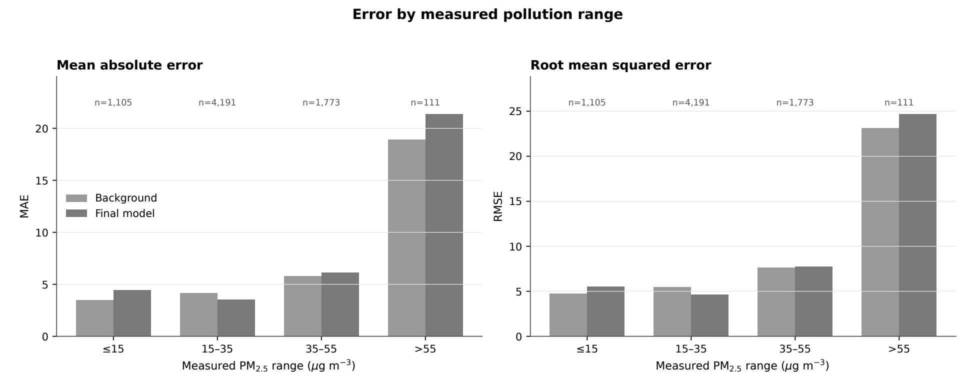


Figure 7: Taiwan concentration-stratified error. The conditioned image improves aggregate performance and the 15–35 μg m^{-3} stratum, but not every concentration range; high-concentration observations remain difficult.

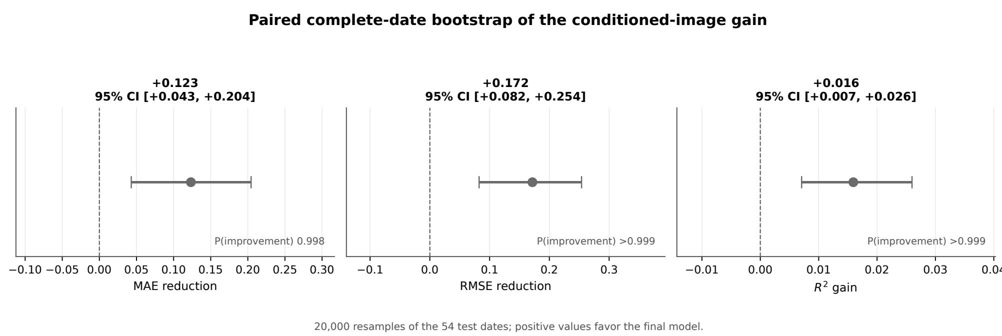


Figure 8: Paired complete-date bootstrap for the frozen Taiwan model relative to the other-site monitoring-network background. Positive reduction denotes lower error. Intervals are percentile 95% intervals from 20,000 resamples of all 54 chronological test dates.

5.3. Complete-date TRAQID evaluation

Table 7 shows the validation ladder. On all 20 dates, the strongest zero-shot LODO configuration combined nonvisual and image context, obtaining MAE 28.53 μg m^{-3}, RMSE 42.14 μg m^{-3} and pooled R^2=0.188. This protocol transition established the need for background conditioning under date isolation. The 16-date reference-assisted model improved MAE to 23.94 $\mu\mathrm{g\,m^{-3}}$, RMSE to 36.23 $\mu\mathrm{g\,m^{-3}}$ and pooled R^2 to 0.361. On the matched future-90% subset, calibration-assisted prediction further increased pooled R^2 from 0.371 to 0.490. The oracle diagnostics reached 0.582 and 0.663, but use the full evaluated date and therefore represent ceilings, not inference-realistic performance. Rows with different coverage are not treated as paired tests.

Pooled skill did not imply stable within-date tracking (Table 6). The 20-date model had mean and median date-wise R^2 of -5.625 and -1.042, with three positive dates; its date-centred R^2 was -0.071. The calibrated future-90% model improved macro date RMSE to 27.80 $\mu\mathrm{g\,m^{-3}}$, but mean date-wise R^2 remained -1.861 and only three of 16 dates were positive. Deterministically retaining every seventh target changed pooled R^2 only from 0.188 to 0.196 for 20-date LODO and from 0.490 to 0.493 for calibration. Thus dense overlap inside an outer-test date does not explain the pooled grouped scores, but between-date baseline variation still contributes materially to them.

Table 6: TRAQID grouped robustness. Macro RMSE averages one value per date; date-centred R^2 removes each date's observed and predicted mean. The non-overlapping-window sensitivity is reported in the surrounding text.

Model	Dates	Macro RMSE	Mean date R^2	Median date R^2	Positive dates	Centred R^2
20-date zero-shot	20	38.73	-5.625	-1.042	3	-0.071
16-date reference-assisted	16	35.89	-4.063	-0.736	1	-0.184
Future 90%, zero-shot	16	34.15	-3.442	-0.667	0	-0.147
Future 90%, calibrated	16	27.80	-1.861	-0.583	3	-0.147

Table 7: TRAQID validation ladder. Future-90% rows form one matched pair; other rows differ in coverage and are not treated as paired comparisons.

Protocol / model	n	MAE	RMSE	Pooled R^2	Status
Random conditional architecture	3,983	5.36	10.29	0.950	Contaminated
20-date MERRA-2 LODO	26,558	28.53	42.14	0.188	Zero-shot
16-date local-reference LODO	20,779	23.94	36.23	0.361	Zero-shot
Future 90%, uncalibrated	18,554	22.67	35.28	0.371	Matched baseline
Future 90%, 10% meta-calibrated	18,554	**21.04**	**31.77**	**0.490**	Assisted[a]

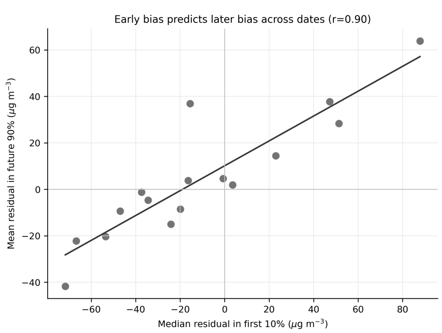
[a]Date-cluster bootstrap 95% interval for pooled R^2: -0.070–0.772. Target-informed oracle ceilings are confined to Appendix B.

5.4. Framework-level performance across mobile and fixed-camera settings

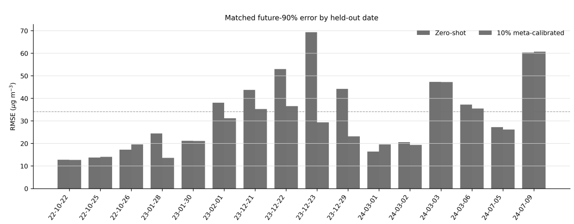
The complete framework combines broad pollution state with learned local scene information rather than asking any one modality to reconstruct the full concentration. On TRAQID, atmospheric and local-reference backgrounds enabled progressively stronger complete-date estimates, culminating in the calibration-assisted future-90% result of R^2=0.490. On Taiwan, the multimodal model improved the prespecified monitoring-network background from R^2=0.690 to 0.706 on the chronological period, while nested held-out-site evaluation retained R^2=0.702 and positive skill at all six sites. These results support the framework as a practical way to combine background, temporal imagery and structured context under different sensing geometries. Complete component ablations and selection sensitivity are retained in Appendix F.

5.5. Chronological calibration-assisted evaluation

Early median residual and later mean residual were strongly associated across the 16 dates (Pearson $r = 0.90$; Figure 9a). Mapping this relationship using other dates reduced matched-test MAE by 7.2% and RMSE by 9.9%, while pooled R^2 increased from 0.371 to 0.490. RMSE improved on 12 of 16 dates; the mean date-level reduction was 6.34 $\mu\mathrm{g\,m^{-3}}$ and the median reduction was 1.11 $\mu\mathrm{g\,m^{-3}}$. The worst-date RMSE decreased from 69.34 to 60.68 $\mu\mathrm{g\,m^{-3}}$. The disparity between mean and median improvement was driven partly by 2023-12-23, whose RMSE decreased by 40.02 $\mu\mathrm{g\,m^{-3}}$. Excluding that date reduced the matched pooled comparison from 0.132 to 0.249 rather than 0.371 to 0.490; the gain persisted but was substantially smaller. This influence analysis and the broad grouped interval make the calibration result evidence of useful date-specific correction, not uniform cross-date accuracy.



(a) Early-to-future residual relation.



(b) Matched RMSE for every date.

Figure 9: Chronological calibration diagnostics. The association in (a) is descriptive; each target-date correction was fitted without that date's future targets.

Table 8: Calibration methods on the same future 90% sequences. Meta-ridge uses other dates to learn how early bias persists.

Calibration	MAE	RMSE	Pooled R^2	Bias
None (zero-shot)	22.67	35.28	0.371	-0.68
Direct first-10% mean offset	27.96	39.48	0.212	-16.52
Direct first-10% median offset	28.85	40.58	0.167	-19.89
Direct first-10% affine OLS	40.89	93.32	-3.404	-22.05
Meta-ridge early mean	21.38	32.11	0.479	-0.91
Meta-ridge early median	**21.04**	**31.77**	**0.490**	-0.91

Naive direct correction failed because the beginning of a drive was often more extreme than its remainder. The cross-date ridge mapping learned to shrink this offset. Date-cluster bootstrap 95% intervals were MAE 16.43–25.99, RMSE 22.72–40.09 and pooled R^2 -0.070–0.772 for the calibrated method; the corresponding zero-shot intervals were MAE 17.92–28.08, RMSE 26.63–43.20 and pooled R^2 -0.201–0.640. The width of these intervals prevents a claim of uniform performance across future dates.

This protocol also requires a functioning roadside $PM_{2.5}$ sensor during the first 10% of each new drive and a contemporaneous external reference during the remainder. It is therefore suitable for online sensor/background calibration or gap filling after warm-up, not sensor-free deployment.

Concentration-stratified analysis exposed the safety-relevant failure mode (Figure 10). Calibration reduced RMSE from 109.75 to 95.96 $\mu g\,m^{-3}$ above 150 $\mu g\,m^{-3}$, but these 921 high-concentration sequences remained far harder than observations below 90 $\mu\mathrm{g\,m^{-3}}$. It also worsened RMSE in the 60–90 $\mu\mathrm{g\,m^{-3}}$ band (22.05 to 25.71 $\mu\mathrm{g\,m^{-3}}$), confirming that a pooled gain did not improve every regime.

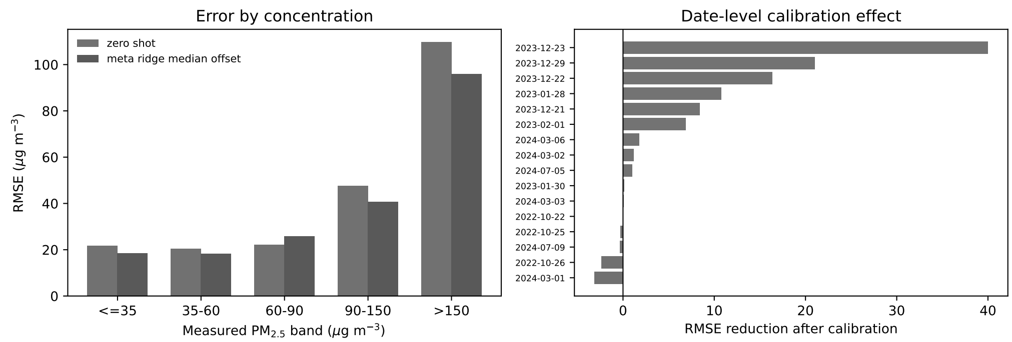


Figure 10: TRAQID robustness analysis. Left: future-90% zero-shot and calibrated RMSE by measured concentration. Right: matched date-level RMSE changes; positive bars favour calibration. No model was refitted or selected for this analysis.

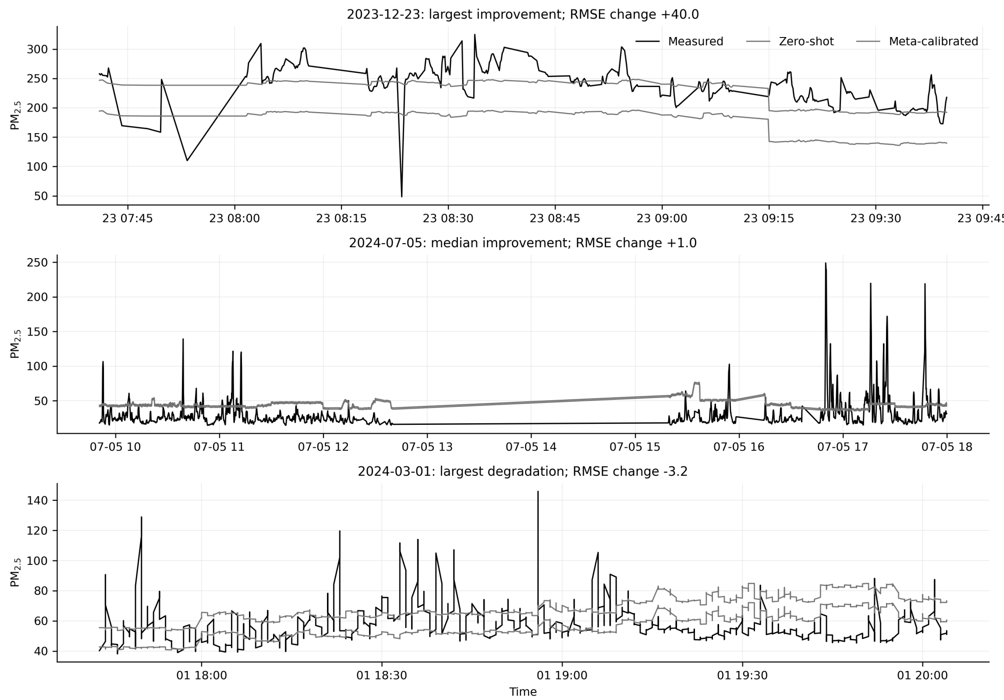


Figure 11: Measured, zero-shot and calibrated time series for the dates with the largest RMSE improvement, median improvement and largest degradation. Dates were selected mechanically from the paired RMSE differences rather than chosen for visual appearance.

6. Discussion

6.1. Validation protocol dominates the apparent result

The difference between random-window R^2=0.950 and 20-date LODO R^2=0.188 is not a small degradation caused by one architectural choice. It is a change in the scientific question. The random protocol asks whether the model interpolates within a drive when almost every target has appeared in training context. LODO asks whether relationships learned on other dates survive a new pollution regime. Reporting only the random score would therefore mischaracterize deployment capability.

The overlap audit complements rather than invalidates prior image-based work. Random-sequence results remain useful for reproduction, debugging and same-distribution interpolation. They become problematic only when interpreted as evidence of future-date generalization.

The constructive implication is that model design and validation design cannot be separated. The random model was implicitly free to recover background state from adjacent observations; the grouped task removes that shortcut. Supplying an inference-time background explicitly and restricting the learned model to a local refinement is therefore a direct response to the failure exposed by the audit, rather than an unrelated increase in model complexity.

The Taiwan results show why the audit should be paired with a constructive evaluation rather than treated only as a critique. With dates isolated before window construction, a background-conditioned model retained R^2=0.706 over 54 retrospectively defined future dates. With complete sites isolated and fusion selected inside training sites, pooled R^2 remained 0.702 and every site score was positive. The two protocols test different shifts, yet both support the same design principle. A strong inference-time network background carries broad spatiotemporal variation, while learned branches estimate a restrained local departure. The fitted estimators themselves are dataset-specific.

6.2. Why monitoring-network background and calibration help

Coarse atmospheric fields cannot reproduce roadside concentration directly, but they identify the broad aerosol regime. A nearby reference station further anchors city-scale conditions. The local model then estimates departure from that anchor. The improvement from reference-only pooled R^2=0.198 to reference-context local modelling at 0.361 confirms that the learned local component remains necessary.

The first 10% of a new date supplies information about sensor/background offset that atmospheric products do not resolve. Directly applying that early offset overcorrected because pollution evolved during the drive. Learning the early-to-later relationship across other dates provided a restrained correction and improved most dates. This is a practical calibration-assisted task, but it requires roadside labels during adaptation and a contemporaneous reference input throughout inference.

The synchronized Taiwan background is conceptually related to CAMS, MERRA-2 and the TRAQID OpenAQ input, but it is local, contemporaneous and multi-site. It is not target leakage because site s is excluded when predicting site s. It does, however, change the deployment requirement: other network monitors must report at the target hour. The one-hour-lag ablation (R^2=0.647) shows that stale observations materially reduce performance, while the single-site outage audit increased background RMSE by at most 0.20 $\mu\text{g}\,\text{m}^{-3}$. The 0.690 background-only score establishes how much predictive information that infrastructure already provides; comparisons against it, not against a global mean, quantify the value of learned context and imagery.

6.3. Interpretation of the multimodal architecture

The framework assigns complementary roles to its inputs. Atmospheric products and synchronized monitors establish the broad pollution state; temporal images encode visibility and persistent scene appearance; structured road, traffic and meteorological variables describe local operating conditions.

The local model then learns a restrained departure from the supplied background rather than reconstructing the entire concentration from imagery alone.

This design was most effective in the fixed Taiwan scenes, where camera geometry was stable and a synchronized network supplied local background conditions. The complete model improved the prespecified background on the chronological test and retained positive performance at every held-out site. Mobile TRAQID posed a harder shift because route, scene, sensor state and regional pollution changed together; there the strongest improvement came from combining reference context with explicit early-date calibration. The two settings therefore demonstrate how a shared governing formulation can be instantiated with separate models according to the available sensing infrastructure.

Component-level and exploratory background comparisons are reported in Appendices Appendix F and Appendix G. They do not replace the prespecified frozen evaluations, but identify useful future extensions: learned spatial background fields, visibility-specific pretraining, sky/road region separation and domain adaptation selected without target-test inspection.

6.4. Limitations and scope

The study has seven central limitations. First, sequences remain autocorrelated within grouped dates even when partitions are leakage-safe. Second, the TRAQID reference-assisted analysis covers 16 dates and one usable OpenAQ station approximately 4.97 km from the campaign centre. Third, its 10% protocol changes the task from zero-shot prediction to calibration-assisted future-period prediction. Fourth, the Taiwan task assumes contemporaneous access to other network sites and is therefore not camera-only estimation. Fifth, both datasets and the retrospective Taiwan test period were available during the broader research programme; a new prospectively frozen campaign is still required for definitive external confirmation. Retrospective network-model extensions are confined to the appendix. Sixth, the redistributed Taiwan files used here do not contain embedded station coordinates, pairwise geometry or monitor instrument identifiers, so those network characteristics could not be audited directly from those files. Seventh, the models estimate concentration, not chemical source contributions; visual, traffic and road features are associative proxies.

Grouped bootstraps reflect date or site sensitivity, but 16 TRAQID dates, 54 Taiwan test dates and six Taiwan sites do not represent all cities, instruments, seasons or camera systems. The grouped intervals and per-group errors should therefore accompany pooled headline values.

7. Conclusion

This study developed and evaluated a leakage-aware multimodal framework that combines an atmospheric or monitoring-network background with temporal imagery and structured local context. On TRAQID, the frame audit showed that the random-window score of R^2=0.950 primarily measured interpolation among overlapping sequences. Under complete-date evaluation, local-reference conditioning achieved zero-shot R^2=0.361, while first-10% calibration-assisted prediction improved the matched future-90% result from R^2=0.371 to 0.490 and reduced RMSE by 9.9%.

On the six-site Taiwan network, the complete estimator achieved R^2=0.706, RMSE 6.44 $\mu\mathrm{g\,m^{-3}}$ and MAE 4.58 $\mu\mathrm{g\,m^{-3}}$ over 7,180 sequences from 54 retrospectively defined future dates, with a positive paired date-bootstrap improvement over the prespecified network background. Nested leave-one-site-out evaluation retained pooled R^2=0.702 and positive R^2 at all six held-out sites. Together, the two datasets demonstrate a flexible framework in which background information captures broad pollution state and learned visual and contextual branches refine the estimate for the local sensing environment.

The broader contribution is a practical, protocol-aware path from shuffled accuracy to credible multimodal estimation: audit raw context, isolate complete dates and sites, exclude self-reference, state which background information is available at inference, and evaluate the full system under the temporal and spatial shifts it is intended to serve. Grouped uncertainty and component sensitivity remain essential qualifications and are reported in the appendices.

Data and software availability

TRAQID and the Taiwan surveillance-image data are publicly available through their source publications (Kathalkar et al., 2024; Wang et al., 2026). Analysis software, experiment configurations, split audits and derived tables are available free of charge at `https://github.com/Kumaryan12/roadside-pm-visual-intelligence`. The exact software and derived-result snapshot used for this manuscript is archived as release `v1.0.1-paper` at Zenodo (doi:10.5281/zenodo.22121572)

(Kumar, 2026). The software is written in Python 3.11 and uses PyTorch, scikit-learn, pandas and NumPy. It runs on macOS or Linux with a CPU; a CUDA-capable GPU is optional for neural-network training. Dependencies are pinned through `pyproject.toml` and `requirements.txt`; primary prediction checksums, executable commands, feature manifests and selection status are listed in Appendix A. The source image datasets are not redistributed because their original licences and access conditions remain controlling; accession links are provided in their cited source publications. OpenAQ observations and global atmospheric products remain subject to their respective providers' access and redistribution terms, and the repository records retrieval and derivation procedures. This combined manuscript supersedes the earlier standalone TRAQID draft, which will not be submitted separately.

Ethics and data governance

This study performed secondary analysis of publicly released research datasets and public environmental observations. No new human participants were recruited and no attempt was made to identify individuals appearing incidentally in mobile-road or fixed-camera imagery. The analysis did not use face or licence plate recognition, identity labels or person-level tracking. Deployment on new imagery would require jurisdiction-specific privacy review, retention limits and, where appropriate, de-identification. Use and redistribution of source data remain governed by the original dataset and service terms.

CRediT authorship contribution statement

Aryan Satyendra Kumar: Conceptualization, Methodology, Software, Validation, formal analysis, visualization, data curation, writing—original draft, and writing—review and editing.

Competing interests

The author declares no competing interests.

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Appendix A. Experiment provenance and selection status

Table A.9: Provenance of headline and diagnostic results. "Frozen" means that model or correction selection used only declared training/validation groups; it does not imply prospective collection.

Result	Primary artifact	Selection status	Permitted interpretation
TRAQID random R^2=0.950	Random conditional run and overlap audit	Diagnostic	Contaminated same-distribution interpolation
TRAQID 20-date LODO R^2=0.188	Multi-source LODO predictions	Outer grouped	Zero-shot complete-date performance
TRAQID 10% calibration R^2=0.490	Chronological calibration predictions and split audit	Outer-safe assisted	Calibrated future-90% performance
TRAQID oracle 0.582–0.663	Oracle residual analysis	Target-informed	Non-operational diagnostic ceiling
Taiwan chronological R^2=0.706	Chronological reference run	Frozen retrospective	Future-period performance on 54 dates
Taiwan LOSO R^2=0.702	Nested conditioned-image report	Nested outer-site	Spatial transfer across six held-out sites
Taiwan learned network 0.740	Chronological network ablation	Post-hoc	Hypothesis for prospective confirmation only

The baseline sensitivity analysis uses `pipelines/china_surveillance_aqi/analyze_reviewer_baselines.py` with the frozen chronological predictions, 5,000 complete-date bootstrap resamples and seed 42. It does not refit the frozen image model. The sequence manifest, prediction tables, split audit and generated summaries are retained with the repository workflow; raw third-party imagery is not redistributed.

The TRAQID robustness analysis is implemented by `pipelines/image_embeddings/analyze_traqid_reviewer_metrics.py`. It reads preserved outer-test predictions only: no model is refitted and no test result is used for selection. The three primary TRAQID prediction files have SHA-256 prefixes `7005fee2e774` (20-date LODO), `7bec2ab0c306` (reference-assisted LODO) and `3212eca9e387` (chronological calibration). The frozen Taiwan chronological prediction file has prefix `362e59a70b67`. Complete hashes and deterministic split manifests are retained beside the run artifacts.

Table A.10: Taiwan monitoring-site metadata and retained availability. Site names and regional context follow the source publication; codes and counts follow the redistributed processed manifest. Coordinates and instrument identifiers were not embedded in the redistributed files.

Dataset code	Site name	Location context	Rows	Dates	First timestamp	Last timestamp
48	Qiaotou	Southern Kaohsiung	5,533	259	2019-07-01 19:00	2020-04-06 20:00
49	Renwu	Southern Kaohsiung	5,952	273	2019-07-01 19:00	2020-04-06 20:00
50	Fengshan	Southern Kaohsiung	3,743	170	2019-10-17 20:00	2020-04-06 20:00
51	Daliao	Southern Kaohsiung	3,765	170	2019-10-17 20:00	2020-04-06 20:00
52	Linyuan	Southern Kaohsiung	5,930	271	2019-07-01 19:00	2020-04-06 20:00
58	Xiaogang	Southern Kaohsiung	3,688	169	2019-10-17 20:00	2020-04-06 20:00

Appendix B. Target-informed TRAQID diagnostic ceilings

The following rows use every target from the evaluated date and are therefore not operational model results. They isolate how much residual error is associated with date-specific offset and scale.

Table B.11: Achieved reference-assisted performance and target-informed ceilings on the same 16 complete dates.

Diagnostic	MAE	RMSE	Pooled R^2	Bias
Zero-shot reference-context model	23.94	36.23	0.361	1.04
Oracle date-mean correction	19.08	29.29	0.582	0.00
Oracle date-affine correction	15.30	26.32	0.663	0.00

Appendix C. Complete per-date calibration errors

Table C.12 reports the matched future-90% errors for every date. No date is omitted. Per-date R^2 is not used to rank models because several dates have narrow target variance, making the statistic unstable; MAE, RMSE and bias are reported consistently for both methods.

Table C.12: Matched future-90% performance by date.

Date	n	Zero MAE	Cal. MAE	Zero RMSE	Cal. RMSE	RMSE change
2022-10-22	1,531	11.25	11.25	12.72	12.72	+0.00
2022-10-25	1,003	10.19	11.23	13.73	14.04	-0.31
2022-10-26	1,965	12.97	14.64	17.19	19.59	-2.39
2023-01-28	993	21.45	10.52	24.42	13.62	+10.79
2023-01-30	1,130	16.76	15.15	21.20	21.05	+0.14
2023-02-01	1,216	24.14	28.45	38.02	31.13	+6.88
2023-12-21	625	29.68	28.51	43.72	35.27	+8.44
2023-12-22	651	44.42	30.31	52.94	36.57	+16.37
2023-12-23	381	65.76	20.90	69.34	29.32	+40.02
2023-12-29	356	37.71	13.51	44.15	23.15	+21.01
2024-03-01	963	11.59	16.10	16.38	19.55	-3.17
2024-03-02	1,118	15.00	11.49	20.56	19.35	+1.20
2024-03-03	1,731	29.77	30.11	47.27	47.19	+0.08
2024-03-06	1,434	21.09	27.82	37.20	35.44	+1.76
2024-07-05	1,894	22.19	20.74	27.21	26.20	+1.01
2024-07-09	1,563	38.59	39.00	60.31	60.68	-0.37

Appendix D. Calibration split audit

Every date used the first chronological segment, kept equal timestamps on one side of the boundary and removed six overlapping sequences. Across 16 dates, 2,129 sequences were used for adaptation, 96 were purged and 18,554 were tested. The raw-row overlap after purging was zero for every date.

Appendix E. Taiwan held-out-site results

Table E.13: Nested leave-one-site-out performance of the conservative conditioned-image model. Image weight α was selected without the outer test site.

Held-out site	n	Selected α	MAE	RMSE	R^2
48	5,006	0.24	5.530	7.228	0.655
49	5,595	0.30	4.409	5.741	0.778
50	3,583	0.30	4.277	5.553	0.735
51	3,619	0.14	5.758	8.022	0.534
52	5,684	0.00	5.073	6.740	0.683
58	3,532	0.06	3.630	4.761	0.767
Pooled	27,019	–	4.818	6.460	0.702

Appendix F. Component ablations and fusion sensitivity

The following component results are retained for complete reporting while the main text emphasizes the prespecified full-framework evaluations. On all 20 TRAQID dates, adding images to the nonvisual MERRA-2 local model increased pooled R^2 from 0.180 to 0.188 and reduced RMSE by 0.19 $\mu\mathrm{g\,m^{-3}}$. On the 16-date reference-covered subset, the reference-context configuration was the strongest tested variant. These differences show that the balance among background, imagery and structured context depends on sensing geometry and background availability.

Table F.14: Component ablation under complete-date TRAQID LODO.

Coverage	Local model	MAE	RMSE	Pooled R^2
20 dates	Nonvisual	28.69	42.34	0.180
20 dates	Nonvisual + images	**28.53**	**42.14**	**0.188**
20 dates	Nonvisual + structured visual	29.63	42.94	0.157
20 dates	Full multimodal	29.31	42.76	0.164
16 reference dates	Reference context	**23.94**	**36.23**	**0.361**
16 reference dates	Reference context + images	25.37	37.65	0.310
16 reference dates	Reference context + structured visual	26.25	38.16	0.291
16 reference dates	Full multimodal	26.13	38.29	0.286

On the frozen Taiwan chronological test, the context-only stage obtained R^2=0.685 compared with 0.690 for the prespecified network background, while the complete conditioned-image model reached 0.706. Under nested LOSO, context improved the background from 0.689 to 0.699 and the complete estimator reached 0.702. The six-site paired interval for the final increment over context included zero; this component-level uncertainty does not alter the achieved performance of the complete estimator but limits claims about any single modality in isolation.

Appendix G. Exploratory Taiwan network-background extension

Table G.15: Exploratory models on the already-inspected Taiwan chronological test period. Selection used validation only and no test target entered fitting, but these values require confirmation on a fresh future period.

Model	MAE	RMSE	R^2	Validation-selected weight
Monitoring-network median	4.704	6.613	0.690	–
Conditioned-image model	4.580	6.442	0.706	0.116
Lagged spatiotemporal background	4.290	6.100	0.736	0.346
Learned spatial background	4.259	6.085	0.738	0.538
Nonnegative ensemble	**4.234**	**6.056**	**0.740**	sums to 1

Appendix H. Protocol-aware literature context

Table H.16: Representative published image-based results. Values are not directly rankable across datasets or split protocols.

Study	Inputs	Split protocol	Target	Reported R^2	Interpretation
Wang et al. (Wang et al., 2024)	Surveillance image sequences	Random two-fold sequence CV	$PM_{2.5}$	0.940	Overlapping-sequence risk
Luo et al. (Luo et al., 2020)	Images + weather/time	Study-specific	$PM_{2.5}$	0.850	Temporal isolation differs
Liu et al. (Liu et al., 2024)	Fixed camera + reference monitor	Chronological study-specific	Hourly $PM_{2.5}$	0.980	Fixed-scene task
PM25-Vision (Han, 2026)	Global street images	Random image split	Daily AQI label	0.550	Different target and scale
TRAQID, this work	T=7 multimodal	Random overlapping windows	Roadside $PM_{2.5}$	0.950	99.90% context leakage
TRAQID, this work	Reference context + 10% adaptation	Complete-date + purged future 90%	Roadside $PM_{2.5}$	0.490	Calibration-assisted
Taiwan, this work	Other-site background + conditioned images	Chronological 54-date test	$PM_{2.5}$	0.706	Frozen retrospective test
Taiwan, this work	Same framework, inner-CV fusion	Nested leave-one-site-out	$PM_{2.5}$	0.702	Six unseen sites

Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this work, the author used OpenAI ChatGPT and Codex to assist with language editing, manuscript organization, code review and the preparation of explanatory figures. The author reviewed and verified all AI-assisted material, analyses, references and figures and takes full responsibility for the content of the publication. No AI-generated primary observational data were used.

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